

GIS Software Developer with 3+ years of experience at MapmyIndia, specializing in custom GIS application development using ArcObjects, C#, VB.NET, and .NET frameworks. Proficient in end-to-end geospatial workflows — spatial database design (PostgreSQL/PostGIS), geoprocessing, map production, and third-party system integration. Experienced in satellite data analysis via Google Earth Engine and Python. M.Tech in Geo-Informatics with research in ML-based landslide inventory mapping. Actively expanding into WebGIS development (Leaflet, OpenLayers, GeoServer) and Geo-AI — applying machine learning to geospatial problem-solving, remote sensing, and environmental analysis.

Experience

MapmyIndia

GIS Software Developer
Delhi

Sep 2022 - Present

- Experience developing custom GIS applications using ArcObjects and the ArcGIS software.
- Expertise in programming languages such as C , VB.NET, .Net which are commonly used to develop ArcObjects applications
- Knowledge of GIS concepts such as spatial data analysis, cartography, and geoprocessing.
- Strong problem-solving and troubleshooting skills, as ArcObjects development often involves complex data and software interactions.
- Familiarity with Esri's development tools, such as ArcMap, ArcScene, and ArcGIS Server.
- Application Query building with PostgreSQL and PostGIS.
- Wrote and developed new and well-tested code for different software projects. Integrated third-party applications into existing software, enhancing features and performance.
- Produced high-quality outputs and geographical information presentations in user-focused tools and displays.
- Maintained current industry knowledge, trends, and regulation updates to inform work.
- Updated new and modified existing applications to client needs and requests. Improved research methodology, developing new statistical methods and models.
- Documentation and Well presentation of work/ workflow.
- Satellite data analysis with Google Earth Engine and Python.

Education

Maulana Abul Kalam Azad University of Technology, WB

M.Tech In Geo -Informatics
9.4

2020 - 2022

Techno International New Town, Kolkata

B.Tech In Civil Engineering
8.46

2016 - 2020

Projects

Flood Analysis Model

PyQGIS · PyQt5 · Python

- Built a desktop GIS application with a custom PyQt5 GUI enabling interactive flood risk zone delineation and spatial analysis workflows.
- Automated QGIS geoprocessing pipelines using PyQGIS scripting for DEM-based hydrological modelling and flood extent mapping.

Public Service Accessibility Analysis

Focus Areas: Hospitals, schools, population coverage, accessibility index| Technologies: PyQGIS, PostGIS, PostgreSQL, Python, Leaflet.js, GeoServer, HTML, CSS, JavaScript

- Performed accessibility analysis for public services such as hospitals and schools.
- Calculated accessibility index, served area, distance to nearest hospitals and schools, and population served.
- Used PostGIS spatial queries for distance-based analysis, service area evaluation, and population coverage assessment.
- Applied Python for statistical reporting, charts, graphs, and analytical summaries.
- Conducted cluster analysis to identify spatial patterns and underserved areas.
- Generated GIS-based reports to support planning and public service decision-making.r

Google Earth Engine Air Quality Change Detection Application

Google Earth Engine, JavaScript, Remote Sensing

- Developed a Google Earth Engine application to monitor and visualize air quality changes (NO2, AOD) using Sentinel-5P satellite time-series data.
- Created visual outputs to compare changes across selected time periods and geographic areas.

Skills

- ArcGIS
- QGIS
- Google Earth Engine
- ArcMap
- ArcScene
- ArcGIS Server
- ArcObjects
- PyQGIS
- PyQt
- ArcPy
- GDAL
- OpenLayers
- Leaflet
- C#
- VB.NET
- .NET
- Python
- JavaScript
- HTML
- CSS
- PostgreSQL
- PostGIS
- Oracle SQL
- GeoServer
- Web-based GIS dashboards
- spatial web applications
- Geoprocessing
- network/accessibility analysis
- served area analysis
- distance analysis
- cluster analysis
- spatial statistics
- Python charts
- graphs
- statistical reports
- map outputs
- geospatial dashboards
- AutoCAD
- Google Earth Engine
- GIS documentation
- workflow presentation
- Critical Thinking

Landslide Inventory Development Using Machine Learning

M.Tech Dissertation | Technologies: GIS, Remote Sensing, Machine Learning

- Developed a landslide inventory using geospatial data and machine learning-based methodology.
- Applied GIS and remote sensing concepts for landslide mapping and spatial data interpretation.
- Worked on data preparation, model-based analysis, and result presentation for geohazard assessment.

Web-Based Public Service Accessibility Dashboard

Technologies: Leaflet, HTML, CSS, JavaScript, WebGIS

- Developed a web-based dashboard for visualizing public service accessibility analysis.
- Displayed hospital and school accessibility results through interactive maps and web layers.
- Built map-based visualization features using Leaflet, JavaScript, HTML, and CSS.
- Designed the dashboard to support easier interpretation of accessibility gaps and service coverage.

Publications

Published research journal at 10th Inter University Engineering, Science & Technology Academic Meet 2019.

FOSET

Certifications

- Artificial Intelligence for Earth Observation and Geodata Handling and Processing - Completion Certificate
- edX Verified Certificate: Introduction to Cloud Development with HTML, CSS, and JavaScript
- Faculty Development Program on Geographical Information System: A Multidisciplinary Tool - Certificate of Participation
- Structural & Foundation Analysis - Online Project-Based Course, September 2020
- Quantity Surveying - Online Project-Based Course, September 2020